

Java Program to Check Whether the Given Number is a Prime Palindrome or Not

Objective / Aim

To write a Java program to check whether the given number is both prime and palindrome.

Algorithm / Logic Description

1.Start the program.

2.Read a number from the user.

3.Check whether the number is prime:

If the number is less than or equal to 1, it is not prime.

4.Check whether the number is divisible by any number from 2 to $\sqrt{n}$.

If divisible, it is not prime.

5.Check whether the number is palindrome:

6.Store the original number.

7.Reverse the digits of the number.

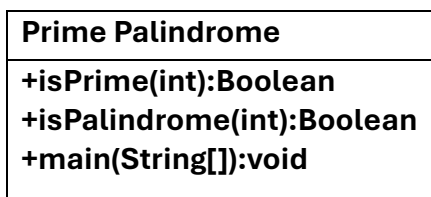
8.Compare the reversed number with the original number.

9.If the number is both prime and palindrome, display Prime Palindrome.

Otherwise, display Not a Prime Palindrome.

9.Stop the program.

UML Class Diagram :



Source Code :

```
import java.util.Scanner;

class PrimePalindrome {

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter a number: ");

int n = sc.nextInt();

// Check Prime

boolean prime = true;

if (n <= 1) {

    prime = false;

}

else {

    for (int i = 2; i <= n / 2; i++) {

        if (n % i == 0) {

            prime = false;

            break;

        }

    }

}

// Check Palindrome

int original = n;

int reverse = 0;

while (n > 0) {

    int digit = n % 10;

    reverse = reverse * 10 + digit;

    n = n / 10;

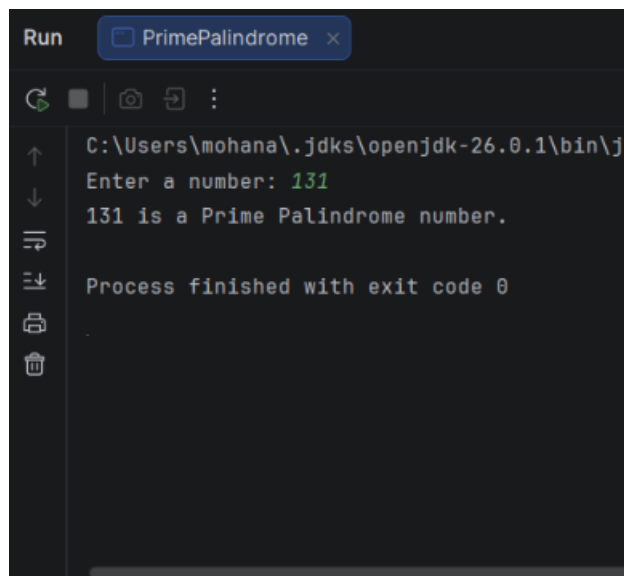
}

if (prime && original == reverse) {

    System.out.println(original + " is a Prime Palindrome number.");
```

```
}  
else {  
    System.out.println(original + " is NOT a Prime Palindrome number.");  
}  
}  
}
```

Output Screen :



```
Run PrimePalindrome x  
C:\Users\mohana\.jdk\openjdk-26.0.1\bin\java  
Enter a number: 131  
131 is a Prime Palindrome number.  
Process finished with exit code 0
```